TruthLens: Knowledge Distillation and INT8 Quantization for Lightweight AI-Generated Image Detection

Nafeesa Mahek (S23108186) and Najeyah Waseem (S23108393)
CS4085 Deep Learning, Effat University
Supervisor: Dr Naila Marir

1 Introduction and Motivation

Image generators can now produce photographs that most people cannot tell apart from real ones, and such images spread quickly online. Our long-term goal is a mobile application that tells users whether an image on their screen is likely to be AI-generated. Running the detector on the phone keeps images private and works offline, but it requires a model that is both small and fast. Since current state-of-the-art detectors contain tens to hundreds of millions of parameters (Chivaran and Ni, 2025), this project investigates how small and efficient a detector can become while remaining accurate, in line with the course theme of Tiny Deep Learning.

2 Related Work

MobileNetV3 (Howard et al., 2019) was designed for fast inference on mobile phone CPUs, whereas ResNet (He et al., 2016) is a larger and widely used image classification backbone. Knowledge distillation (KD) trains a small student network to imitate the softened predictions of a larger teacher network (Hinton et al., 2015). Integer quantization stores weights and activations as 8-bit integers, reducing model size and speeding up inference on mobile hardware (Jacob et al., 2018).

For AI-generated image detection, GenImage (Zhu et al., 2023) provides real ImageNet photographs together with fake images from eight different generators. LAID (Chivaran and Ni, 2025) benchmarked lightweight models on a subset of GenImage and found that MobileNetV3-Small reaches 95.19% accuracy. However, LAID applied neither distillation nor quantization, did not measure real inference latency, and names knowledge distillation as future work. Tiny-LaDeDa (Cavia et al., 2024) showed that distillation can improve a very small detector (from 87.4% to 91.6% mAP), but on a different dataset and without quantization.

2.1 Research Gap

None of these studies combines knowledge distillation and INT8 quantization for a MobileNetV3-Small detector on GenImage while also measuring real inference latency. Our project investigates this combination under fair, controlled conditions.

3 Research Question

Does knowledge distillation from a ResNet-50 teacher improve the accuracy of a MobileNetV3-Small AI-generated image detector on GenImage, and is this improvement preserved after the distilled model is quantized to INT8?

4 Proposed Approach

4.1 Baseline and Teacher

The baseline M0 is an ImageNet-pretrained MobileNetV3-Small that we fine-tune on our dataset using only the true labels. The teacher T is an ImageNet-pretrained ResNet-50 fine-tuned on the same training data. Both models classify each image into nine classes (real, or one of the eight generators), which gives the teacher richer soft targets than a real-versus-fake task. The application reports $P(\text{AI}) = 1 - P(\text{real})$.

4.2 Deep Learning Contribution (M1)

M1 uses the same architecture, starting weights and training budget as M0, so that the only difference between them is the training strategy. M1 is trained with the distillation loss of Hinton et al. (2015):

$$\mathcal{L} = \alpha\,\mathrm{CE}(y, p_s) + (1-\alpha)\,T^2\,\mathrm{KL}\big(p_t^{(T)} \,\|\, p_s^{(T)}\big),$$

where p_s and p_t are the student and teacher predictions, $p^{(T)}$ is the softmax at temperature T, and α balances the true labels against the teacher's guidance; both are tuned on the validation set.

4.3 Efficiency Contribution (M2)

M2 is obtained by converting M1 to 8-bit integers. We first apply post-training quantization,

and if this causes too large an accuracy drop, we use quantization-aware training (QAT), as recommended by Jacob et al. (2018) for small networks.

5 Hypotheses

5.1 H1 (Deep Learning)

M1 will achieve at least 1.5 percentage points higher test accuracy than M0, averaged over three random seeds, and this gain will be larger than the seed-to-seed standard deviation of either model.

5.2 H2 (Efficiency)

M2 will be at least three times smaller and at least 30% faster than M1 (median latency, batch size one, on an INT8-capable laptop or phone CPU), while retaining at least half of M1's accuracy gain over M0. If M1 does not outperform M0, we will compare M2 with M0 directly.

5.3 Basis for the Targets

Both targets are exploratory. H1 is motivated by the headroom above LAID's 95.19%, the distillation gain in Tiny-LaDeDa, and our pilot (80.3% after two epochs). The H2 size target matches our pilot (3.3–3.5×), and the speed target follows Jacob et al. (2018).

6 Dataset and Experiments

6.1 Dataset

We build a balanced subset of GenImage following LAID's protocol. The training set contains 100,000 images (6,250 real and 6,250 fake images per generator) taken from GenImage's training split, and GenImage's validation split is divided equally into a validation set and a test set of 16,000 images each. All images are resized to 256×256 and saved as JPEG at quality 95. This removes a shortcut, since GenImage's real images are JPEG and its fakes PNG (Cavia et al., 2024); random JPEG compression is also applied during training. Sampling uses a fixed seed, and the file lists are saved on GitHub.

6.2 Experiments

E1 compares M1 with M0, E2 compares M2 with M1, and E3 compares M2 with M0. E4 is an ablation study that varies the distillation temperature ($T \in \{2, 4, 8\}$) and compares a nine-class teacher with a two-class teacher. All hyperparameters are selected on the validation set, and the test set is used only once, for the final evaluation. Quality is measured by accuracy, macro-F1, AUC, per-generator accuracy and accuracy on JPEG-compressed and blurred images; cost is measured by parameters, file size, MACs, median latency with its interquartile range, and peak memory.

Measurement	Value
Time per epoch	1.22 min
Peak GPU memory	0.63 GB
Val. accuracy / F1 / AUC	80.3% / 0.80 / 0.90
Parameters / MACs	1.53 M / 61.5 M
Size FP32 → INT8	6.12 → 1.8 MB
CPU latency (FP32, Xeon)	2.20 ms

Table 1: Pilot results for M0 (2 epochs, seed 42).

7 Feasibility

7.1 Pilot Results

We trained M0 for two epochs on a balanced subset of Tiny-GenImage, a public scaled-down version of GenImage, using 9,375 training and 1,875 validation images on a T4 GPU (Table 1). The model passed the overfit-one-batch test. An early post-training INT8 test cut the model size by 3.3–3.5×, but accuracy collapsed to about 47% and latency rose, as the Colab CPU lacks INT8 (VNNI) support; this shaped our M2 plan (Section 8). Scaling up suggests about 13 minutes per M0 epoch on 100,000 images with free Kaggle and Colab GPUs.

7.2 Team Roles

We are a two-person team approved by the instructor. Nafeesa leads the proposal, M1 and M2 experiments, latency and the Flutter demo app; Najeyah leads the dataset analysis, teacher and M0 training, and the final paper. Both review each other's work and share the writing.

7.3 Weekly Plan (to early December)

Week	Tasks
1	Build dataset; submit proposal (Item 1)
2	Dataset checks; train the teacher
3	Train and evaluate M0 (Item 2)
4	Distillation (M1); quantization (M2)
5	Seeds, ablations, latency (Item 3)
6	Test evaluation, error analysis, paper, poster (Item 4)

8 Risks and Backup Plan

- **INT8 accuracy collapse** (seen in the pilot): use QAT, keep sensitive layers in FP32, or fall back to FP16.
- **No INT8 speed-up on Colab**: also measure latency on an INT8-capable laptop or phone.
- **Small distillation gain**: report it honestly and analyse it with the ablations.
- **Compute limits**: use a half-size 66,000-image subset built the same way.

AI-Use Statement

We used AI assistants (Claude and Perplexity) to summarise papers, compare ideas, debug code, and help draft and edit this proposal. The authors chose the research direction, ran the pilot, and verified all references and numbers. The AI self-review is in Appendix A.

References

Bar Cavia, Eliahu Horwitz, Tal Reiss, and Yedid Hoshen. 2024. Real-time deepfake detection in the real-world. *Preprint*, arXiv:2406.09398.

Nicholas Chivaran and Jianbing Ni. 2025. LAID: Lightweight AI-generated image detection in spatial and spectral domains. In *Proceedings of the International Conference on Privacy, Security and Trust (PST)*. ArXiv:2507.05162.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *Preprint*, arXiv:1503.02531. NIPS 2014 Deep Learning Workshop.

Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. 2019. Searching for MobileNetV3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1314–1324.

Benoit Jacob, Skirmantas Kligys, Bo Chen, Menglong Zhu, Matthew Tang, Andrew Howard, Hartwig Adam, and Dmitry Kalenichenko. 2018. Quantization and training of neural networks for efficient integer-arithmetic-only inference. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2704–2713.

Mingjian Zhu, Hanting Chen, Qiangyu Yan, Xudong Huang, Guanyu Lin, Wei Li, Zhijun Tu, Hailin Hu, Jie Hu, and Yunhe Wang. 2023. GenImage: A million-scale benchmark for detecting AI-generated image. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, volume 36.

A AI Review Report

Self-review using the course AI Reviewer prompt (formative feedback only; the instructor assigns the official grade).

Criterion	Score
Motivation	9/10
Related work (at least 5 papers)	9/10
Research question and gap	9/10
Deep learning contribution	14/15
Efficiency contribution	13/15
Two hypotheses	9/10
Dataset and fair experiments	10/10
Feasibility and team plan	9/10
Risks and fallback	5/5
Writing, format, AI disclosure	5/5
Total	**92/100**

Readiness

READY TO SUBMIT (no major issues). M0, M1 and M2 are clearly defined; H1 compares M1 with M0, while H2 compares M2 with M1 and checks M2 against M0. Both hypotheses have numerical targets, and the test set is reserved for the final evaluation.

Three Main Improvements

(1) Report seed variation alongside mean accuracy in later items. (2) Keep the quantization fallback (QAT or FP16) ready, since the pilot showed that post-training INT8 is fragile. (3) Measure latency on the same device for all models.

Major and Minor Issues

Major issues: none. Minor issues: the timeline is compact, and distillation gains may be small and should be interpreted together with the ablation results.

References to Verify

The venue details for LAID (PST 2025) and the preprint status of Tiny-LaDeDa.

Questions the Instructor May Ask

Why should distillation help on a near-binary task? Why can a smaller INT8 model be slower? How is a fair comparison between M0 and M1 ensured?